

MAZEN WAEI

Software Engineer | Backend & Full-Stack | AI/ML

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EDUCATION

B.Sc. Software Engineering — Computer Science 2022 – Jun 2026

Cairo University Cairo, Egypt

Grade: Very Good

TECHNICAL SKILLS

Languages: C++, Python, C#, Java, JavaScript (ES6+), TypeScript

Web & Frameworks: .NET, ASP.NET Core, React, Flask, REST API Design, WebSocket

Databases: MySQL, SQL Server, FAISS, Query Optimization, Schema Design

DevOps & Tools: Git, GitHub Actions (CI/CD), Docker, Nginx, Linux

AI & ML: Retrieval-Augmented Generation (RAG), LangChain, TensorFlow, scikit-learn, Vector Embeddings, Pandas

Training / Intern

Software Engineering Intern

Sep 2023 – Nov 2024

ALX Africa Cairo, Egypt

- Completed a rigorous, 14-month intensive software engineering training program focused on backend architecture, system design, and DevOps.
- Collaborated within a 3-engineer Agile team to architect and deploy a capstone full-stack marketplace platform (Market Mate).
- Developed backend RESTful APIs utilizing Flask and MySQL, applying systematic query indexing and N+1 query elimination to optimize response times.
- Gained hands-on DevOps experience by automating CI/CD pipelines via GitHub Actions and deploying containerized Linux applications using Docker and Nginx.

PROJECTS

Market Mate — Full-Stack Marketplace Platform [Github](#) Flask • MySQL • Docker • GitHub Actions

- Designed and built a full-stack marketplace with JWT-based authentication, a product catalog serving 50+ SKUs, and real-time buyer/seller notifications via WebSocket. Deployed on Linux with Nginx and Docker.
- Reduced API response latency by 30% through systematic MySQL indexing and N+1 query elimination; improved system reliability with a CI/CD pipeline that cuts deployment overhead by 60%.

Egyptian Legal AI — RAG-based Legal Assistant (Graduation Project) [Github](#) Python • Fast API • Lang Chain • Gemini • React

- Architected a Retrieval-Augmented Generation (RAG) system for Egyptian Labor Law, delivering accurate Arabic legal responses with cited article references using FAISS vector search,
- LangChain
- Built a production FastAPI backend with SQL Server chat history and JWT authentication, handling concurrent sessions with sub-5s response latency and a 100% pass rate across 75 comprehensive test cases.

Airbnb Clone — Distributed Booking API [Github](#) Flask • MySQL • HTML/CSS

- Built a RESTful API in Flask handling booking and listing workflows across 5 normalized resource models; implemented full CRUD operations with a responsive HTML/CSS frontend and SQL schema enforcement.

Heart Disease Prediction — ML Classification Model [Github](#) Python · scikit-learn · Matplotlib · Seaborn

- Trained logistic regression and random forest classifiers on the UCI Heart Disease dataset; achieved 87% classification accuracy after hyperparameter tuning with GridSearchCV.
- Conducted exploratory data analysis across 13 clinical features and produced 8 diagnostic visualizations to surface feature correlations and class imbalances.

Habit Tracker App — ASP.NET Core CRUD Application [Github](#) C# · ASP.NET Core · SQL Server · EF Core

- Designed EF Core database models and implemented full CRUD controllers with Razor views in ASP.NET Core MVC, delivering a functional personal productivity tool with persistent SQL Server storage.

CERTIFICATIONS

Full Stack Software Engineering <i>ALX Africa</i>	Sep 2023 – Nov 2024
Deep Learning Specialization <i>DeepLearning.AI (Coursera)</i>	Jun 2025 – Oct 2025
AI & Machine Learning Program <i>Microsoft / Sprints</i>	Jun 2025 – Nov 2025
EF SET English Certificate — C2 Proficient (71/100) <i>EF SET</i>	Mar 2025